Quantitative Sales Forecasting

Exam code: 9BS0

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Introduction to quantitative sales forecasting

- The sales forecast is an important **business planning tool**
  - It provides an **estimation of future sales figures** using **past data** and considering predictable **external factors**
- Sales forecasts can be used to **identify trends in product sales**, which can then be compared with the market as a whole

Methods used in quantitative sales forecasting

Method	Explanation
Moving averages	• A series of averages calculated from **successive segments of a series of data** • These averages **smooth data** so that trends may be more easily identified
Extrapolation	• The prediction of future sales from past data • Extrapolation can often be done simply by extending a **line of best fit**
Correlation	• Where there is **a link between two variables**, there is a correlation • **Correlations** may be positive or negative

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

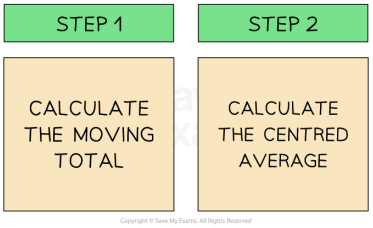
Calculation of time: Series analysis

Moving averages

- Sometimes, **past sales data is too erratic** for clear trends to be identified

- Four-month or twelve-month moving averages are used where **seasonality** is a key factor in sales

Calculating moving averages



Steps to calculate the moving average

- The moving total is calculated by **adding together sales figures for a specified number of periods**
  - E.g. a **three-month moving total** is calculated by **adding the first three months**, **followed** by the second three months and the third three months and so on
- The centred average is calculated by **dividing the moving total by the specified number of periods**
  - E.g. a **three-month centred average** is calculated by **dividing the three-month moving total by three**
- A **series of centred averages** is known as the **moving average**

Worked Example

RJ Inflatables is a manufacturer of novelty celebration balloons. Its monthly sales from January to July are shown in Table A.

Rachel Jameson, the managing director, is concerned that sales are declining but is struggling to identify a trend with the sales data she has available. Rachel's financial administrator has suggested using a moving average so that she can forecast future sales with greater accuracy.

Table A

January	42,000
February	51,000
March	45,000
April	33,000
May	47,000
June	30,000
July	45,000

Calculate a three-month moving average using RJ Inflatables's January to July sales data.

(6)

Step 1: Calculate the three-month moving total of sales for each group of three months

Month	Sales (£)	Three-month Moving Total	
January	42 000	–	
February	51 000	–	
March	45 000	138 000	JANUARY + FEBRUARY + MARCH
April	33 000	129 000	FEBRUARY + MARCH + APRIL
May	47 000	125 000	MARCH + APRIL + MAY
June	30 000	110 000	APRIL + MAY + JUNE
July	45 000	122 000	MAY + JUNE + JULY

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Step 2: Calculate the three-month centred average for each group of three months

Month	Sales (£)	Three-month moving total	Three-month centred-average	
January	42,000			
February	51,000		46,000	138,000 / 3
March	45,000	138,000	43,000	129,000 / 3
April	33,000	129,000	41,667	125,000 / 3
May	47,000	125,000	36,667	110,000 / 3
June	30,000	110,000	40,667	122,000 / 3
July	45,000	122,000		

- When plotted on the same graph, the three-month centred average provides a **smoother curve**, which makes **extrapolation** of the data for **forecasting** relatively straightforward

A comparison of raw sales data and 3-month centred average data

Interpreting scatter graphs

- Scatter graphs allow businesses to compare **two variables**, such as sales volume and advertising, to establish if there is any **correlation** between them

Example scatter graph

An example of a scatter graph showing the number of sales managers employed by a business and the volume of items sold

Types of correlation

- A correlation exists where there is a **relationship** or **connection** between two variables
  - A **positive correlation** means that **as one variable increases**, **so does the other variable**
    - A **line of best fit** that **slopes upwards** can be identified
  - A **negative correlation** means that **as one variable increases**, **the other variable decreases**

between the two variables
    - Therefore, it is not possible to identify **a line of best fit**

Diagrams of correlation types

The main types of correlation between two variables: positive, negative and no correlation

- Correlation does not always indicate a relationship or **causation** between two sets of variables, so businesses need to **conduct research** to establish **whether a relationship exists**, as well as the **strength of that relationship**
- Where a **line of best fit can be identified**, and when **causation** is determined, a business can **extrapolate** the data to make **predictions** around changes to either of the variables
    - E.g. **extrapolating** the line of best fit in the example below, the business could predict that **employing seven sales managers would likely result in sales of 46 units**
- Extrapolation assumes that what has **happened in the past** will be the same as what will happen in the future

Extrapolation using a line of best fit

An example of a scatter graph with a line of best fit showing the number of sales managers employed by a business and the volume of items sold

Examiner Tips and Tricks

Watch out for outlying data — if there is more than one outlier above the line, adjust your line of best fit upwards. Similarly, if there is more than one outlier below the line, adjust your line of best fit downwards. Just one outlier should not influence your line of best fit.

Limitations of quantitative sales forecasting

- Quantitative sales forecasting techniques are useful when the future is expected to reflect what has happened in the past
- It is especially effective in the **short term**
- In the longer term, uncertainties — especially in the **external environment** — can make simple extrapolation of past data unreliable

Factors affecting the accuracy of sales forecasts

Examples of external factors that may influence the accuracy of the sales forecast include changes to legislation and market changes

- In many cases, the **sales forecast** can provide little more than an estimate of future performance
  - As long as it is **approximately accurate**, businesses can **use the sales forecast to plan resources**, such as staff, finance and production, and to **produce budgets**
- Businesses can improve the accuracy of sales forecasts by:
  - Conducting detailed **market research**
  - Employing experts with excellent **market knowledge**
  - **Revising** the sales forecasts frequently

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